[Description]

GCFLU Quadrivalent Pre-filled Syringe inj. is a vaccine containing colorless or slightly whitish liquid made by splitting and inactivating influenza virus cultured by inoculating the allantoic cavity of embryonated egg in order to maintain antigenicity.

[Composition]

1 Pre-filled Syringe 0.5ml contains,

Active Ingredient: Purified inactivated influenza virus antigen (In-house) ····· 60 μg

A/Missouri/11/2025 IVR-279 (H1N1) ..... 15 μg

A/Michigan/105/2025 SAN-049A (H3N2) 15μg

B/Tokyo/EIS13-175/2025..... 15μg

B/Phuket/3073/2013 ····· 15 μg

Buffer: Sodium chloride (EP) ..... 4 mg

Potassium chloride (EP)..... 0.1 mg

Disodium hydrogen phosphate dihydrate (EP)..... 0.6 mg

Potassium dihydrogen phosphate (EP)····· 0.1 mg

Diluent: Water for Injection (EP) ..... q.s.

Needle (Sterilized disposable needle (25G x 5/8 (0.5 x 16 mm) ..... 1 ea

This vaccine complies with the WHO recommendations (Northern Hemisphere) for 2026-2027 season.

[Indications]

Prophylaxis against influenza caused by influenza A subtype viruses and type B viruses in persons aged 6 months and older.

[Dosage & Administration]

An intramuscular injection of the following dose and immunization of one dose is necessary in every year at same volume. Aged 6 months and older: A single dose of 0.5 ml

The children younger than 9 years of age who have not been vaccinated should be vaccinated two doses at an interval of at least 4 weeks.

The preferred sites for intramuscular injection are the anterolateral aspect of the thigh (or the deltoid muscle of the upper arm if muscle mass is adequate) in children 6 through 35 months of age, or the deltoid muscle of the upper arm in children from 36 months of age and adults.

The safety and efficacy of the vaccine was not established in children younger than 6 months.

[Precautions for use]

1. Contraindications

Examine subjects by history taking and visual inspection and if necessary, by auscultation and percussion. Vaccination is prohibited when subject is diagnosed as one of the following cases. However, if there is a possibility of influenza infection and it is determined that there is no concern for significant disorder due to vaccination, subject may be vaccinated.

1) Person with severe hypersensitivity to the following components
   - Component of this vaccine (including active ingredients and excipients)
   - Egg proteins (ovalbumin), formaldehyde
2) Person who showed severe hypersensitivity reactions (e.g., anaphylaxis) following previous administration of influenza vaccine

2. Take special care

1) As with other vaccines administered by intramuscular injection, this vaccine should be given with caution in individuals with thrombocytopenia or any coagulation disorder since bleeding may occur following intramuscular administration.

2) In person who showed Guillain-Barre syndrome or neurological disorders within 6 weeks from the previous influenza vaccination, the benefits and risks of vaccination should be carefully considered.

3) Vaccination should be postponed in person suffering from severe febrile illness or acute illness.

3. Adverse Reactions

1) There is possibility of local reactions such as redness, swelling and pain, or systemic reactions such as fever, chills, headache, fatigue and vomiting. But they usually disappear within 2-3 days.

2) In rare cases, acute disseminated encephalomyelitis (ADEM) may occur.

Fever, headache, convulsions, dyskinesia and consciousness disorder usually occur within 2 weeks following the administration of the vaccine. When these symptoms are suspected, appropriate medical treatment should be available by diagnosis with MRI and so on.

3) Allergic reaction or anaphylactic shock may occur in very rare cases.

4) Transient disorders of systemic and local nervous system may rarely occur. Palsy, neuralgia, cerebral hemorrhage or inflammation of the nervous system (ex: Guillain-Barre syndrome) have been reported.

5) Safety of the vaccine was evaluated for the 4 clinical studies performed with healthy children, adults, and elderly. In children aged 6 through 35 months who received the vaccine, 115

subjects (67.6%) out of 170 subjects showed adverse events. Adverse drug reactions were 82 subjects (48.2%) and no serious adverse drug reactions were reported. In children aged 3 through 18 years who received the vaccine, 218 subjects (68.3%) out of 319 subjects showed adverse events. Adverse drug reactions were 204 subjects (63.9%) and no serious adverse drug reactions were reported. In adults aged 19 through 64 years who received the vaccine, 415 subjects (71.2%) out of 583 subjects showed adverse events. Adverse drug reactions were 399 subjects (68.4%) and no serious adverse drug reactions were reported.

In elderly over 65 years of age who received the vaccine, 148 subjects (43.8%) out of 338 subjects showed adverse events.

Adverse drug reactions were 140 subjects (41.4%) and no serious adverse drug reactions were reported.

(1) Solicited adverse drug reactions within 7 days of vaccination are listed in the table below.

		Children aged 6 through 35 months (n=170)	Children aged 3 through 18 years (n=319)	Adults aged 19 through 64 years (n=583)	Elderly over 65 years of age (n=338)
Local	Pain	27.6%	52.7%	48.9%	21.0%
	Tenderness		54.5%	56.8%	27.5%
	Erythema/Redness	11.8%	6.6%	7.9%	3.8%
	Induration/ Swelling	5.9%	8.2%	5.8%	3.6%
Systemic	Drowsiness[1)]	15.9%	-	-	-
	Fever	6.5%	3.1%	0.9%	0.3%
	Sweating	2.4%	2.2%	4.3%	2.7%
	Chills	2.4%	5.0%	7.7%	4.4%
	Nausea/ vomiting	2.4%	0.6%	2.2%	0.9%
	Diarrhea	5.9%	0.3%	1.5%	1.2%
	Fatigue	-	15.4%	25.6%	10.7%
	Malaise	-	11.0%	7.5%	8.3%
	Headache	0.6%	6.9%	13.4%	7.1%
	Muscle aches	7.6%	8.2%	26.4%	6.5%
	Arthralgia	-	1.6%	5.8%	3.6%

1) Drowsiness only applies for children aged 6 months through 35 months

(2) Unsolicited adverse drug reactions occurring within 28 days or 21 days of vaccination were reported in 4 subjects (2.4%) from children aged 6 through 35 months (Infections and infestations: 3 subjects, Skin and subcutaneous tissue disorders: 1 subject), 3 subjects (0.9%) from children aged 3 through 18 years (General disorders and administration site conditions: 2 subjects, Infections and infestations: 1 subject), 13 subjects (2.2%) from adults Infections and infestations: 5 subjects, investigations: 2 subjects, Respiratory thoracic and mediastinal disorders: 2 subjects, Musculoskeletal and connective tissue disorders: 1

subject, Nervous system disorders: 1 subject, Skin and subcutaneous tissue disorders: 1 subject, General disorders and administration site conditions: 2 subjects), and 4 subjects

{1.2%) from elderly Infections and infestations: 1 subject, General disorders and administration site conditions: 1 subject, investigations: 1 subject, Nervous system disorders: 1 subject)

(3) Serious adverse events occurring within 6 months of vaccination were reported in 13 subjects (7.6%) from children aged 6 through 35 months (Pneumonia: 4 cases, Influenza: 3 cases , Bronchitis: 2 cases , Pneumonia respiratory syncytial viral: 1 case, Bronchiolitis: 1 case, Croup infectious: 1 case, Gastroenteritis norovirus: 1 case, Gastroenteritis rotavirus: 1 case, Urinary tract infection: 1 case, Gastrointestinal infection: 1 case, Impaired healing: 1 case, Foreign body in gastrointestinal tract: 1 case, Febrile convulsion: 1 case), 5 subjects (1.6%) from children aged 3 through 18 years (Pharyngitis: 1 case, Headache: 1 case, Mesenteric lymphadenitis: 1 case, Acute gastroenteritis: 1 case, Peritonsillar Abscess: 1 case, Acute appendicitis: 1 case), 5 subjects (0.9%) from adults (Cystitis: 1 case, Pulmonary Tuberculosis: 1 case, Breast mass: 1 case, ileus : 1 case, Gastric cancer: 1 case), and 4 subjects (1.2%) from elderly (Pain: 1 case, Arthralgia: 1 case, Herpes zoster: 1 case, Gastric cancer: 1 case), but they were evaluated as 'not related' to the product.

6) Results of post-marketing surveillance in South Korea

(1) The results of post-marketing surveillance conducted domestically for 4 years on 2,060 adult subjects aged 19 years and older in order to go through a re-examination showed that the incidence of adverse events was 10.49% (216 out of 2,060 subjects, 578 cases in total, regardless of causal relationship. Among these, no serious adverse events and serious adverse drug reactions have been reported. In addition, unexpected adverse events and unexpected adverse drug reactions are listed in the following table according to their frequency of onset.

		Unexpected Adverse Events Regardless of Causal Relationship 2.33% (48 out of 2,060 subjects, 72 cases)	Unexpected Adverse Drug Reactions of Which Causal Relationship Cannot Be Ruled Out 0.29% (6 out of 2,060 subjects, 7 cases)
Rarely (≥ 0.01% and < 0.1%)	Respiratory, thoracic, and mediastinal disorders	Asthma	Cough
	Gastrointestinal disorders	Dyspepsia, benign gastrointestinal neoplasm, gastrointestinal disorder NOS, hemorrhoids	-
	General disorders and administration site condition	Injection site inflammation	Injection site inflammation
	Nervous system disorders	Apa thy, insomnia, dizziness, cerebral ischemia.	Apathy, dizziness
	Eye disorders	Blepharitis, conjunctivitis	-

	General disorders and administration site condition	Back pain	-
	Vascular disorders	Hypertension	-
	Cardiac disorders	Palpitation	Palpitation
	Metabolism and nutrition disorders	Hyperlipidemia	-
	Infections and infestations	Fungal dermatitis, moniliasis	-
Uncommonly (≥ 0.1% and < 1%)	Respiratory, thoracic, and mediastinal disorders	Rhinitis, sinusitis, cough, upper respiratory tract infection	-
	Gastrointestinal disorders	Gastritis, gastroesophageal reflux, abdominal pain, irritable bowel syndrome	-
	Skin and subcutaneous tissue disorders	Dermatitis, pustular rash, contact dermatitis, urticaria	-
	General disorders and administration site condition	Cellulitis, injection site pruritus	Injection site pruritus

(2) The results of post-marketing surveillance conducted domestically for 4 years on 2,033 pediatric subjects aged ≥ postnatal 6 months and < 19 years showed that the incidence of adverse events was 30.74% (625 out of 2,033 subjects, 1,221 cases in total), regardless of causal relationship. Among these, serious adverse events and serious adverse drug reactions are listed in the following table according to their frequency of onset.

		Serious adverse Events 0.10% (2 out of 2,033 subjects; 2 cases)	Serious adverse Events 0.00% (0 out of 2,033 subjects; 0 cases)
Rarely (≥ 0.01% and < 0.1%)	White cell and reticuloendothelial system disorders	Kawasaki disease*	-
	Respiratory system disorders	Bronchitis	-

* Unexpected serious adverse event

In addition, unexpected adverse events and unexpected adverse drug reactions are listed in the following table according to their frequency of onset.

	Unexpected Adverse Events Regardless of Causal Relationship 7.85% (159 out of	Unexpected Adverse Drug Reactions of which Causal Relationship Cannot

		2,033 subjects; 178 cases)	Be Ruled Out 0.49% (10 out of 2,033 subjects; 10 cases)
Rarely (≥ 0.01 % and < 0.1%)	Application site disorders	Cellulitis, Injection site bruising	Injection site bruising
	Body as a whole-general disorders	Leg pain, Influenza-like symptoms, Hypothermia, Temperature changed sensation	Leg pain, Hypothermia, Temperature changed sensation
	Gastrointestinal system disorders	Constipation, Gastroesophageal reflux	-
	Skin and appendages disorders	Acne, dermatitis contact, Dermatitis fungal, Skin disorder	Rash pustular
	Resistance mechanism disorders	Moniliasis	Otitis media
	White cell and reticuloendothelial system disorders	Kawasaki disease, Lymphadenopathy	-
	Secondary terms-events	Varicella	-
Uncommonly (≥ 0.1% and < 1%)	Respiratory system disorders	Sinusitis, Cough, Asthma	-
	Application site disorders	Injection site pruritus	Injection site pruritus
	Gastrointestinal system disorders	Abdominal pain, Stomatitis	-
	Skin and appendages disorders	Dermatitis, Rash pustular, Urticaria, Pruritus.	Urticaria
	Resistance mechanism disorders	Otitis media	-
	Vision disorders	Conjunctivitis	-
Commonly (≥ 1% and < 10%)	Respiratory system disorders	Rhinitis	-

(3) Adverse events from domestic post-marketing surveillance and spontaneously reported data on side effects were comprehensively assessed at the end of post-marketing surveillance along with the adverse events data (1989 to March 31, 2020) reported for all drugs that have been licensed for domestic marketing. Among the adverse events that were reported more frequently with statistical significance for this drug than the adverse events reported for all other drugs, following adverse events were newly identified. However, these results do not mean that the causal relationship between the relevant

ingredient and the following adverse events has been demonstrated.

- Systemic and injection site adverse events: Injection site inflammation, injection site warmth, injection site pruritus, injection site bruising
- Infection: Rhinitis (rhinorrhea)

To report any side effect(s):
• Egypt:
- Egyptian Pharmacovigilance Centre (EPVC):

- e-mail for reporting: **pv.followup@edaegypt.gov.eg**
- Website for reporting: **https://vigiflow-eforms.who-umc.org/eg/med**
- Hotline: 15301
- Scan QR code:

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4. General Cautions

1) Advise the subjects or their guardians that the subjects take a rest on the day and the next day of vaccination keeping the injection site clean. Symptoms of high fever, convulsion appear after vaccination, they should consult a physician quickly.

2) Antibody response in patients with endogenous or iatrogenic immunosuppression may be insufficient.

3) Influenza vaccine should be administered before prevalent. Vaccination can be delayed according to epidemiological situation.

4) Influenza vaccine should be administered with current-year recommended strains.

5. Drug Interactions

1) There is no data or study on co-administration of this product with other vaccines.

2) The immunological response may be diminished if the patient is undergoing immunosuppressant treatment.

3) Following influenza vaccination, false positive results in serologic tests using the ELISA method to detect antibodies against HIV-1, Hepatitis C, and especially HTLV-1 have been observed (The Western Blot technique disproves the false-positive ELISA test results). These transient false-positive results could be due to the IgM response by the vaccine.

6. Use in Pregnancy and Nursing Mothers

1) Pregnancy

- Inactivated Influenza vaccine (egg-derived) is known that it can be used in aft pregnancy cycles regardless of the pregnancy stage. There are more safety data for second trimester and third trimester compared with the first trimester. In addition, according to data on the usage of inactivated influenza vaccine collected globally, no adverse effects of the vaccine on the fetus and maternity were reported.

- In addition, no direct or indirect adverse effects related to reproductive toxicity and developmental toxicity were observed in animal studies conducted using this vaccine. However,

clinical trials have not evaluated the safety of the pregnant women when administered this vaccine.

2) Nursing Mothers

- Inactivated Influenza vaccine (egg-derived) is known that it can be used to lactating women. Restricted data indicate that the vaccine is not known whether the product is excreted in human milk. However, there is no adequate study of vaccination in animals during lactation, and clinical trials have not evaluated the safety of nursing mothers when administered this vaccine.

7. Precautions in Administration

1) Before use check this product visually for particles or discoloration. If either is present, do not use.

2) The injection site is usually lateral upper arm and disinfected with ethanol or tincture of iodine. Repeated injections at the same site should be avoided.

3) Intravenous administration is prohibited.

4) The tip of needle should not penetrate blood vessel.

5) Do not mix with other vaccines in same syringe.

6) Pre-filled syringes are disposable and should not be reused.

8. Precautions for Handling

1) Store at 2-8°C without freezing.

2) The vaccine should be shaken well and mixed homogeneously before use.

[Storage and Shelf life]

Store at 2-8°C without freezing in hermetic container and protect from light.
The shelf life is 12 months from the date of manufacture.

[How Supplied]

0.5 mL/ Pre-filled Syringe x In-house packing unit.

[Revision date]

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